

Project Genesis

Emergence Validation & Telemetry Console

EXPERIMENT: EMERGENCE_SOCIAL_BONDS_ACCELERATED



Run Metadata

Seed: 1720

Scarcity: 3.0

Ticks: 100200

Commit: da209a4

Runtime: 2026-07-18 14:51:29

Unit Tests

PASS

Pass rate: 55/55 passed.

Simulated Population (Select an Agent)

 Agent #0	 Agent #1	 Agent #2	 Agent #3	 Agent #4	 Agent #5
 Agent #6	 Agent #7	 Agent #8	 Agent #9	 Agent #10	 Agent #11
 Agent #12	 Agent #13	 Agent #14	 Agent #15	 Agent #16	 Agent #17
 Agent #18	 Agent #19	 Agent #20	 Agent #21	 Agent #22	 Agent #23
 Agent #24	 Agent #25	 Agent #26	 Agent #27	 Agent #28	 Agent #29
 Agent #30	 Agent #31	 Agent #32	 Agent #33	 Agent #34	 Agent #35
 Agent #36	 Agent #37	 Agent #38	 Agent #39	 Agent #40	 Agent #41
 Agent #42	 Agent #43	 Agent #44	 Agent #45	 Agent #46	 Agent #47
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 Agent #60	 Agent #61	 Agent #62	 Agent #63	 Agent #64	 Agent #65
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 Agent #72	 Agent #73	 Agent #74	 Agent #75	 Agent #76	 Agent #77

 Agent #78	 Agent #79	 Agent #80	 Agent #81	 Agent #82	 Agent #83
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 Agent #1056	 Agent #1057	 Agent #1058	 Agent #1059	 Agent #1060	 Agent #1061
 Agent #1062	 Agent #1063	 Agent #1064	 Agent #1065	 Agent #1066	 Agent #1067



Agent

#103

Profile:

SEDENTARY BUILDER

STATUS

Dead

CAUSE OF DEATH (PRIMARY / SECONDARY)

Dehydration (Secondary: Exposure)

BIOLOGICAL LIFESPAN (AGE / MAX)

12.6 / 90.0 years (Sp:1170, Su:1140, Au:1080, Wi:1158)

LIFE STAGE & GENERATION

undefined (Generation 2 | Colony: Beta)

FAMILY LINEAGE

Parents: [4, 48] | Children: 0

SPAWN BIOME

Unknown

MET POPULATION

69 agents [4, 5, 18, 25, 29, 35, 37, 42, 45, 48, 51, 52, 56, 57, 63, 64, 68, 73, 79, 80, 88, 93, 95, 97, 98, 99, 102, 106, 110, 114, 115, 116, 119, 122, 123, 124, 126, 129, 131, 133, 136, 137, 139, 141, 145, 146, 148, 149, 150, 152, 153, 157, 158, 159, 160, 162, 166, 167, 170, 171, 173, 175, 178, 180, 182, 183, 185, 186, 189]

PHYSIOLOGICAL RESERVES (FAT / MUSCLE)

Fat: 100.0% | Muscle: 100.0%

STORED RESOURCES (FOOD / WATER)

Food Stockpile: 16.4 | Water Stockpile: 1.1

TRAUMA / INJURY LEVEL

38.2% (Injured)

SHELTER STATUS

Primitive tent at (597, 556) | Durability: 90%

EXPLORATION RADIUS & HOME DRIFT

60 cells / undefined cells

TELEMETRY (EATS / DRINKS)

29 eats / 62 drinks

RESTING & MEMORY CHURN

100 ticks / 0 rediscoveries

Behavioral Entropy (Action Distribution)





Exposure and Death History Logs

```
[MILESTONE] Tick 1 (Year 0, Day 1): Complete shelter network achieved – all 16 living agents have shelter!  
[BIRTH] Tick 5 | Colony Alpha | Parents #0 x #1 -> Child #16 (Gen 1)  
[BIRTH] Tick 5 | Colony Alpha | Parents #2 x #3 -> Child #17 (Gen 1)  
[MILESTONE] Tick 5 (Year 0, Day 5): First successful reproduction! (parents #0 x #1) Birth #2.  
[BIRTH] Tick 7 | Colony Beta | Parents #5 x #6 -> Child #18 (Gen 1)  
Year 0, Day 8: Agent 0 upgraded shelter to Wood Cabin at coordinate (146, 503).  
[BIRTH] Tick 10 | Colony Beta | Parents #4 x #7 -> Child #19 (Gen 1)
```

Run History & Comparison (Last 5 Runs)

TIMESTAMP	EXPERIMENT	SCARCITY	SURVIVORS	AVG RADIUS	AVG DISCOVERIES	TESTS
2026-07-18 14:51:29 (Current)	emergence_social_bonds_accelerated	3.0	162/201	28.3 cells	203.4	PASS
2026-07-18 11:51:27	past_rerun_1	5.0	0/24	12.7 cells	36.2	PASS
2026-07-18 12:16:03	emergence_social_bonds_accelerated	3.0	0/16	12.8 cells	23.1	PASS
2026-07-18 13:00:37	emergence_social_bonds_accelerated	3.0	162/201	28.3 cells	203.4	PASS
2026-07-18 13:52:57	emergence_social_bonds_accelerated	3.0	162/201	28.3 cells	203.4	PASS
2026-07-18 14:51:28	emergence_social_bonds_accelerated	3.0	162/201	28.3 cells	203.4	PASS



Comparative Study Module

Overlay past simulation runs as dashed comparison curves to observe selection shifts.

 Executive Summary

Run Summary

Duration:	100000 ticks (277.8 years)
Peak Population:	235 agents
Final Population:	216 alive
Maximum Generation:	Gen 3
Longest Lineage:	309 descendants
Oldest Agent:	115.7 years
Genetic Diversity:	30.4%
Extinct Colonies:	1
Dominant Colony:	Gamma
Dominant Death Cause:	Old Age
Dominant Gene Drift:	Gene 12 (-28%)

Civilization Health Score

POPULATION 92%	ECOLOGY 68%	GENETICS 61%
COGNITION 50%	REPRODUCTION 100%	OVERALL 74%

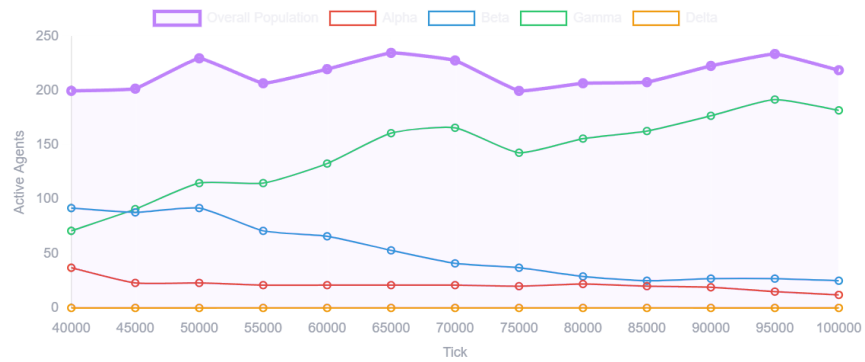
Research Conclusions

- Civilization population stabilized at 215 ± 11 agents after Tick 60000.
- Colony Gamma dominated reproduction, accounting for 349% of all births.
- Learning Horizon gene underwent strong positive selection, drifting -8%.
- Average agent lifespan decreased by 79% from founders to descendants.
- Mortality patterns shifted: primary cause of death transitioned from Old Age to Starvation.

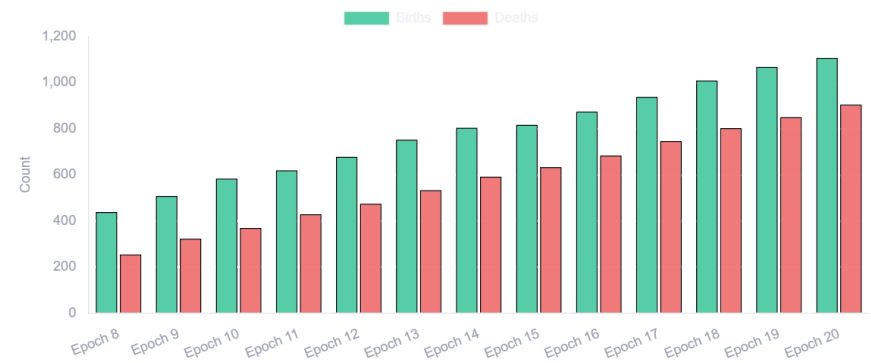


Population Dynamics

Demographic Growth History



Births & Deaths Census Dynamics



Population Growth Ledger

Historical time-series comparing population levels across colonies (corresponds to population.csv).

TICK	YEAR	DAY	TOTAL ALIVE	COLONY ALPHA	COLONY BETA	COLONY GAMMA	COLONY DELTA
40000	Year 111	Day 40	200	37	92	71	0
45000	Year 125	Day 0	202	23	88	91	0
50000	Year 138	Day 320	230	23	92	115	0
55000	Year 152	Day 280	207	21	71	115	0
60000	Year 166	Day 240	220	21	66	133	0
65000	Year 180	Day 200	235	21	53	161	0
70000	Year 194	Day 160	228	21	41	166	0
75000	Year 208	Day 120	200	20	37	143	0
80000	Year 222	Day 80	207	22	29	156	0
85000	Year 236	Day 40	208	20	25	163	0
90000	Year 250	Day 0	223	19	27	177	0
95000	Year 263	Day 320	234	15	27	192	0
100000	Year 277	Day 280	219	12	25	182	0

 Colony Comparison

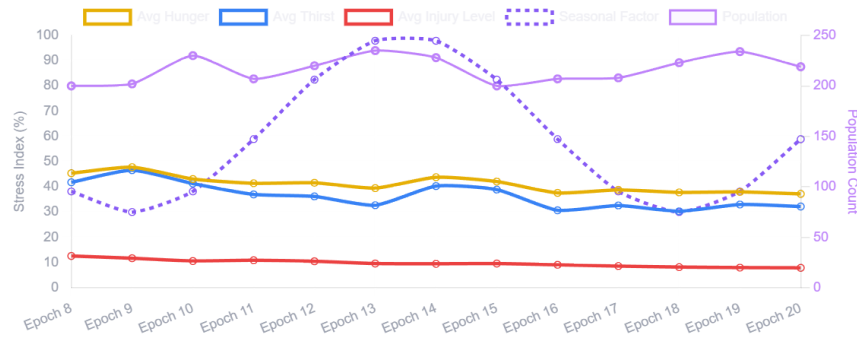
Colony Comparison & Fate Matrix

Comprehensive overview of the 4 independent colonies to evaluate regional survival success factors (corresponds to colonies.csv).

COLONY	STATUS	LIFESPAN (TICKS)	EXTINCTION TICK	PEAK POP	CURRENT POP	BIRTHS	DEATHS	AVG LIFESPAN	HIGHEST GEN	AVG SHELTER	AVG PREDICTOR ACCURACY	FOOD STABILITY	W ST
 Colony Alpha	SURVIVED	15384	—	37	11	+158	-151	52.0 y	Gen 10	Tier 1.0	50.0%	51%	47
 Colony Beta	SURVIVED	15384	—	92	25	+307	-286	51.4 y	Gen 12	Tier 1.0	50.0%	46%	63
 Colony Gamma	SURVIVED	15384	—	192	180	+641	-465	51.1 y	Gen 14	Tier 1.0	50.0%	71%	71
 Colony Delta	EXTINCT	8400	8400	6	0	+2	-6	15.2 y	Gen 1	Tier —	50.0%	51%	45

 Ecology stress

Ecology Stress & Resource Abundance



Ecology & Physiological Status

Analysis of population physiological health relative to scarcity index and resource competition.

AVERAGE HUNGER LEVEL

24.7%

AVERAGE THIRST LEVEL

39.2%

AVERAGE FAT RESERVES

100.0%

AVERAGE MUSCLE MASS

100.0%

Water Economy & Pathfinding Audit

Search and pathfinding metrics for finding fresh water sources.

WATER SEARCH COUNT

39403

AVERAGE PATH EFFICIENCY

99.3%

AVG TIME TO HYDRATION

1.1 ticks

SEARCH SUCCESS RATE

39.7%

Dehydration Deaths Telemetry

Causal variables present at the time of dehydration deaths.

CAUSAL CONTEXT	DEATHS COUNT	SHARE %	STATUS ALERT
Carrying Water at death	7	33.3%	● Normal (Intake delay / loop)
Beside Water source	5	23.8%	● Normal (Reachability / stuck)
Had water locations in memory	21	100.0%	● Critical (Pathing failure)
Visited dried source recently	7	33.3%	● Normal (Drought / seasonality)



Reproduction & Family Trees

Birth & Reproduction Analysis

Generational intervals, children distribution, and spawning dynamics (corresponds to births.csv).

COLONY	TOTAL BIRTHS	AVG GENERATION INTERVAL	BIRTH RATE (PER YEAR)	BIOMES SPAWN DISTRIBUTION	AVG CHILDREN PER ADULT	MAX DESCENDANTS LINEAGE
Alpha	+158	11502 ticks	3.70 / year	Unknown: 158, Temperate Forest: 4	3.0 children	17 children
Beta	+307	11816 ticks	7.18 / year	Unknown: 307, Grassland/Savanna: 4	3.5 children	25 children
Gamma	+641	11492 ticks	15.00 / year	Unknown: 641, Grassland/Savanna: 4	2.7 children	14 children
Delta	+2	89 ticks	0.05 / year	Tundra: 4, Unknown: 2	1.0 children	1 children

Reproduction Observability Funnel

Causal conversions from fertile age potential to successful birth.

STAGE	TICKS / COUNT	CONVERSION %	STATUS ALERT
1. Potential Opportunities (Fertile Age Ticks)	116221	100.0%	● Active
2. Biologically Eligible (Actual Opportunities)	24728	21.3%	● Low Health/Reserve
3. Mate Exists in World Ticks	24728	100.0%	● Normal
4. Mate inside Search Radius Ticks	24647	99.7%	● Normal
5. Acceptable Relationship Ticks	24647	100.0%	● Healthy
6. Mate was Biologically Eligible Ticks	14084	57.1%	● Normal
7. Planner Chose Reproduction Action Ticks	6110	43.4%	● Normal

Reproduction Terminal Failure Partition

Mutually exclusive terminal outcomes of fertile ticks (sums to 100%).

OUTCOME REASON	TICKS / COUNT	SHARE %
Successful Child Birth	370	0.2%
Biological Rejection (Cooldown)	178324	78.8%
Biological Rejection (Low Health)	700	0.3%
Biological Rejection (Low Fat Reserves)	627	0.3%
Biological Rejection (Hunger / low food)	2735	1.2%
Biological Rejection (Thirst / low water)	1548	0.7%
Biological Rejection (Low Shelter Quality)	1056	0.5%
Biological Rejection (Injury Level)	871	0.4%
Unreachable Partner (Out of Range)	162	0.1%
Mate was Biologically Ineligible	21126	9.3%
Utility Gating (Other Priorities)	7974	3.5%
Mate Unwilling (Mate chose other action)	9619	4.3%

STAGE	TICKS / COUNT	CONVERSION %	STATUS ALERT

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Reproduction Action Loss Details

When reproduction had utility but lost, which action took priority?

WINNING ACTION	LOSS COUNTS	PERCENTAGE
Store Water	3096	38.8%
Deposit Water	1705	21.4%
Sheltering	1433	18.0%
Exploring	453	5.7%
Drinking	402	5.0%
Resting	342	4.3%
Eating	142	1.8%
Store Food	134	1.7%
Deposit Food	109	1.4%
Building Shelter	70	0.9%
Drink Stored Water	44	0.6%
Eat Stored Food	27	0.3%
Withdraw Water	15	0.2%

Reproduction Utility by Age Bracket

Average reproduction utility vs. winning utility by age bracket.

BRACKET	DECISIONS	AVG REPRO UTILITY	AVG WIN UTILITY
Juvenile (< 14 years)	267267	0.00	23.83
Young Adult (14-30 years)	70695	4.05	29.68
Mature Adult (30-60 years)	45526	3.00	37.07
Elder (60+ years)	9608	0.00	23.80



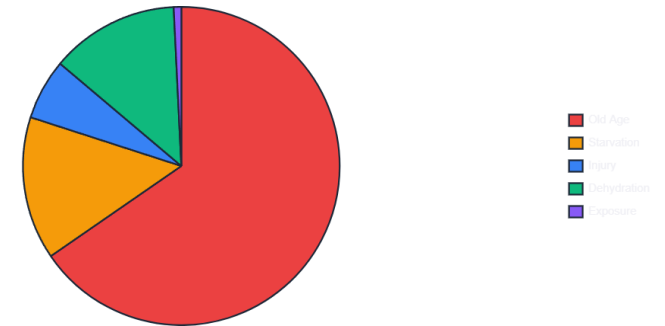
Mortality

Cause of Death Breakdown

Empirical list of mortality reasons across the population (corresponds to deaths.csv).

PRIMARY CAUSE	SECONDARY CAUSE	TOTAL COUNT	PERCENTAGE
Old Age	Natural Decline	286	31.5%
Old Age	Dehydration	255	28.1%
Starvation	Dehydration	67	7.4%
Old Age	Exposure	49	5.4%
Dehydration	None	40	4.4%
Dehydration	Exposure	40	4.4%
Starvation	None	36	4.0%
Injury	Dehydration	28	3.1%
Starvation	Exposure	26	2.9%
Dehydration	Starvation	23	2.5%
Injury	Exposure	17	1.9%
Dehydration	Injury	16	1.8%
Injury	None	11	1.2%

Mortality Pie Chart



PRIMARY CAUSE	SECONDARY CAUSE	TOTAL COUNT	PERCENTAGE
Exposure	Injury	6	0.7%
Old Age	Injury	4	0.4%
Starvation	Iniurv	3	0.3%



Cognitive Learning Indicators

Empirical measurements of predictor trust, concept development, and neural modeling (corresponds to cognition.csv).

COLONY	AVG PREDICTION ACCURACY	AVG PREDICTION TRUST	TOTAL CONCEPTS	TOTAL PROCEDURES
Alpha	50.0%	0.0%	0 concepts	17 procedures
Beta	50.0%	0.0%	0 concepts	20 procedures
Gamma	50.0%	0.0%	0 concepts	4 procedures
Delta	50.0%	0.0%	0 concepts	28 procedures

Spatial Exploration & Memory

Measurements of travel distances, discoveries, knowledge retrieval, and cache hits.

COLONY	TOTAL DISCOVERIES	AVG TRAVEL DISTANCE	AVG KNOWN NODES	AVG CACHE HITS
Alpha	0 items	66.7 cells	0.0 nodes	0.0 hits
Beta	0 items	43.1 cells	0.0 nodes	0.0 hits
Gamma	0 items	17.0 cells	0.0 nodes	0.0 hits
Delta	0 items	50.1 cells	0.0 nodes	0.0 hits



Evolution & Gene Drift

Expressed Gene Selection Matrix

Tracks direction of selection pressure across standard genes (corresponds to genes.csv).

Gene / Trait Name	Founder Mean (Early)	Selection Mean (Late)	Final Variance	Selection Trend
Metabolism	0.6608	0.7160	0.0267	↑ +0.0552
Thermoregulation	0.4126	0.2192	0.0485	↓ -0.1934
Vision	0.6525	0.7467	0.0350	↑ +0.0942
Mobility	0.5199	0.4039	0.0859	↓ -0.1160
Memory Fidelity	0.4981	0.3571	0.0441	↓ -0.1410
Planning	0.3889	0.2668	0.0501	↓ -0.1221
Novelty Seeking	0.1406	0.1277	0.0088	→ -0.0129
Social Proximity	0.5129	0.4369	0.0421	↓ -0.0760
Aggression	0.3511	0.2212	0.0266	↓ -0.1299
Resource Sharing	0.4489	0.3892	0.1206	↓ -0.0597
Risk Sensitivity	0.4591	0.5811	0.0285	↑ +0.1220

Gene / Trait Name	Founder Mean (Early)	Selection Mean (Late)	Final Variance	Selection Trend
Resilience	0.4449	0.7045	0.0717	↑ +0.2596
Longevity	0.6301	0.3477	0.0798	↓ -0.2824
Learning Rate	0.3191	0.0953	0.0297	↓ -0.2238



Evolutionary Success Lineages

Longest Surviving Lineages

RANK	LINEAGE FOUNDER	COLONY	GEN
#1	Agent #1116	● Gamma	Gen 14
#2	Agent #1120	● Gamma	Gen 14
#3	Agent #1122	● Gamma	Gen 14
#4	Agent #1123	● Gamma	Gen 14
#5	Agent #1020	● Gamma	Gen 13

Largest Lineages (Descendants)

RANK	PARENT AGENT	COLONY	CHILDREN
#1	Agent #4	● Beta	25
#2	Agent #29	● Beta	20
#3	Agent #5	● Beta	19
#4	Agent #28	● Alpha	17
#5	Agent #51	● Beta	17

Cognitive Mastery Lineages

RANK	COGNITIVE AGENT	COLONY	ACCURACY
#1	Agent #0	● Alpha	—
#2	Agent #1	● Alpha	—
#3	Agent #2	● Alpha	—
#4	Agent #3	● Alpha	—
#5	Agent #4	● Beta	—



Raw Event Log & Timeline

Interactive Chronological Timeline

Filterable log of all simulation timeline entries (corresponds to timeline.csv).

MILESTONE

Complete shelter network achieved — all 16 living agents have shelter!

Tick 1 (Year 0, Day 1)

BIRTH

Child #16 (Gen 1) born in Colony Alpha to Parents #0 and #1.

Tick 5 (Year 0, Day 5)

BIRTH

Child #17 (Gen 1) born in Colony Alpha to Parents #2 and #3.

Tick 5 (Year 0, Day 5)

MILESTONE

First successful reproduction! (parents #0 x #1) Birth #2.

Tick 5 (Year 0, Day 5)

BIRTH

Child #18 (Gen 1) born in Colony Beta to Parents #5 and #6.

Tick 7 (Year 0, Day 7)

**BIRTH**

Tick 10 (Year 0, Day 10)

Child #19 (Gen 1) born in Colony Beta to Parents #4 and #7.

**MILESTONE**

Tick 10 (Year 0, Day 10)

New population peak reached: 20 alive at tick 10!

**BIRTH**

Tick 23 (Year 0, Day 23)

Child #20 (Gen 1) born in Colony Gamma to Parents #8 and #10.

**BIRTH**

Tick 53 (Year 0, Day 53)

Child #21 (Gen 1) born in Colony Gamma to Parents #9 and #11.

**BIRTH**

Tick 83 (Year 0, Day 83)

Child #22 (Gen 1) born in Colony Delta to Parents #13 and #15.

**BIRTH**

Tick 94 (Year 0, Day 94)

Child #23 (Gen 1) born in Colony Delta to Parents #12 and #14.

**BIRTH**

Tick 366 (Year 1, Day 6)

Child #24 (Gen 1) born in Colony Alpha to Parents #1 and #2.

 BIRTH

Tick 368 (Year 1, Day 8)

Child #25 (Gen 1) born in Colony Beta to Parents #5 and #6.

 BIRTH

Tick 388 (Year 1, Day 28)

Child #26 (Gen 1) born in Colony Gamma to Parents #8 and #10.

 BIRTH

Tick 453 (Year 1, Day 93)

Child #27 (Gen 1) born in Colony Gamma to Parents #9 and #11.

 MILESTONE

Tick 501 (Year 1, Day 141)

Population crisis — dropped below 50 (28 alive). Survival at risk.

 DEATH

Tick 723 (Year 2, Day 3)

Agent #23 died at coordinate (552, 304). Primary: Exposure. Secondary: Injury.

 BIRTH

Tick 727 (Year 2, Day 7)

Child #28 (Gen 1) born in Colony Alpha to Parents #1 and #2.

 BIRTH

Tick 729 (Year 2, Day 9)

Child #29 (Gen 1) born in Colony Beta to Parents #5 and #6.

 BIRTH

Tick 820 (Year 2, Day 100)

Child #30 (Gen 1) born in Colony Gamma to Parents #9 and #10.

 MILESTONE

Tick 820 (Year 2, Day 100)

New population peak reached: 30 alive at tick 820!

 BIRTH

Tick 1183 (Year 3, Day 103)

Child #31 (Gen 1) born in Colony Gamma to Parents #9 and #11.

 DEATH

Tick 1232 (Year 3, Day 152)

Agent #7 died at coordinate (597, 556). Primary: Injury. Secondary: Exposure.

 DEATH

Tick 1518 (Year 4, Day 78)

Agent #19 died at coordinate (600, 569). Primary: Injury. Secondary: Exposure.

 BIRTH

Tick 1942 (Year 5, Day 142)

Child #32 (Gen 1) born in Colony Beta to Parents #4 and #5.

 BIRTH

Tick 2085 (Year 5, Day 285)

Child #33 (Gen 1) born in Colony Gamma to Parents #9 and #11.

**BIRTH**

Tick 2096 (Year 5, Day 296)

Child #34 (Gen 1) born in Colony Alpha to Parents #0 and #2.

**BIRTH**

Tick 2305 (Year 6, Day 145)

Child #35 (Gen 1) born in Colony Beta to Parents #4 and #5.

**BIRTH**

Tick 2468 (Year 6, Day 308)

Child #36 (Gen 1) born in Colony Alpha to Parents #0 and #2.

**DEATH**

Tick 2620 (Year 7, Day 100)

Agent #15 died at coordinate (590, 273). Primary: Exposure. Secondary: Injury.

**BIRTH**

Tick 2763 (Year 7, Day 243)

Child #37 (Gen 1) born in Colony Beta to Parents #4 and #5.

**BIRTH**

Tick 3124 (Year 8, Day 244)

Child #38 (Gen 1) born in Colony Beta to Parents #5 and #6.

**BIRTH**

Tick 3278 (Year 9, Day 38)

Child #39 (Gen 1) born in Colony Alpha to Parents #2 and #3.

 DEATH

Tick 3308 (Year 9, Day 68)

Agent #17 died at coordinate (147, 503). Primary: Injury. Secondary: Exposure.

 BIRTH

Tick 3485 (Year 9, Day 245)

Child #40 (Gen 1) born in Colony Beta to Parents #4 and #5.

 BIRTH

Tick 3640 (Year 10, Day 40)

Child #41 (Gen 1) born in Colony Alpha to Parents #2 and #3.

 DEATH

Tick 3835 (Year 10, Day 235)

Agent #32 died at coordinate (592, 552). Primary: Dehydration. Secondary: Exposure.

 BIRTH

Tick 3847 (Year 10, Day 247)

Child #42 (Gen 1) born in Colony Beta to Parents #4 and #5.

 BIRTH

Tick 4058 (Year 11, Day 98)

Child #43 (Gen 1) born in Colony Alpha to Parents #1 and #2.

 BIRTH

Tick 4179 (Year 11, Day 219)

Child #44 (Gen 1) born in Colony Alpha to Parents #0 and #3.

 BIRTH

Tick 4208 (Year 11, Day 248)

Child #45 (Gen 1) born in Colony Beta to Parents #5 and #6.

 MILESTONE

Tick 4208 (Year 11, Day 248)

New population peak reached: 40 alive at tick 4208!

 DEATH

Tick 4406 (Year 12, Day 86)

Agent #36 died at coordinate (132, 481). Primary: Dehydration. Secondary: Exposure.

 BIRTH

Tick 4498 (Year 12, Day 178)

Child #46 (Gen 1) born in Colony Alpha to Parents #1 and #2.

 BIRTH

Tick 4570 (Year 12, Day 250)

Child #47 (Gen 1) born in Colony Beta to Parents #4 and #5.

 DEATH

Tick 4832 (Year 13, Day 152)

Agent #11 died at coordinate (846, 594). Primary: Dehydration. Secondary: None.

 DEATH

Tick 5186 (Year 14, Day 146)

Agent #6 died at coordinate (597, 556). Primary: Injury. Secondary: Exposure.

 BIRTH

Tick 5540 (Year 15, Day 140)

Child #48 (Gen 1) born in Colony Beta to Parents #4 and #5.

 BIRTH

Tick 5547 (Year 15, Day 147)

Child #49 (Gen 2) born in Colony Gamma to Parents #10 and #26.

 MILESTONE

Tick 5547 (Year 15, Day 147)

First grandchild born — the civilization now spans three generations!

 DEATH

Tick 5672 (Year 15, Day 272)

Agent #3 died at coordinate (146, 503). Primary: Injury. Secondary: None.

 BIRTH

Tick 5762 (Year 16, Day 2)

Child #50 (Gen 1) born in Colony Alpha to Parents #0 and #2.

 BIRTH

Tick 5820 (Year 16, Day 60)

Child #51 (Gen 2) born in Colony Beta to Parents #25 and #29.

 BIRTH

Tick 5901 (Year 16, Day 141)

Child #52 (Gen 1) born in Colony Beta to Parents #4 and #5.

 BIRTH

Tick 5913 (Year 16, Day 153)

Child #53 (Gen 2) born in Colony Gamma to Parents #10 and #21.

 DEATH

Tick 5957 (Year 16, Day 197)

Agent #22 died at coordinate (552, 304). Primary: Exposure. Secondary: Injury.

 DEATH

Tick 5997 (Year 16, Day 237)

Agent #38 died at coordinate (595, 555). Primary: Starvation. Secondary: Dehydration.

 BIRTH

Tick 6008 (Year 16, Day 248)

Child #54 (Gen 2) born in Colony Alpha to Parents #16 and #28.

 BIRTH

Tick 6139 (Year 17, Day 19)

Child #55 (Gen 2) born in Colony Gamma to Parents #9 and #26.

 BIRTH

Tick 6187 (Year 17, Day 67)

Child #56 (Gen 2) born in Colony Beta to Parents #25 and #29.

 BIRTH

Tick 6263 (Year 17, Day 143)

Child #57 (Gen 1) born in Colony Beta to Parents #4 and #5.

 DEATH

Tick 6427 (Year 17, Day 307)

Agent #2 died at coordinate (145, 503). Primary: Starvation. Secondary: Dehydration.

 BIRTH

Tick 6573 (Year 18, Day 93)

Child #58 (Gen 2) born in Colony Gamma to Parents #9 and #20.

 BIRTH

Tick 6626 (Year 18, Day 146)

Child #59 (Gen 1) born in Colony Beta to Parents #4 and #5.

 DEATH

Tick 6670 (Year 18, Day 190)

Agent #39 died at coordinate (129, 519). Primary: Starvation. Secondary: None.

 BIRTH

Tick 6782 (Year 18, Day 302)

Child #60 (Gen 2) born in Colony Gamma to Parents #10 and #30.

 BIRTH

Tick 6845 (Year 19, Day 5)

Child #61 (Gen 2) born in Colony Alpha to Parents #24 and #28.

 BIRTH

Tick 6982 (Year 19, Day 142)

Child #62 (Gen 2) born in Colony Gamma to Parents #9 and #26.

 BIRTH

Tick 6987 (Year 19, Day 147)

Child #63 (Gen 2) born in Colony Beta to Parents #5 and #29.

 MILESTONE

Tick 6987 (Year 19, Day 147)

New population peak reached: 50 alive at tick 6987!

 DEATH

Tick 6990 (Year 19, Day 150)

Agent #34 died at coordinate (145, 502). Primary: Starvation. Secondary: Exposure.

 BIRTH

Tick 7142 (Year 19, Day 302)

Child #64 (Gen 2) born in Colony Beta to Parents #4 and #25.

 BIRTH

Tick 7149 (Year 19, Day 309)

Child #65 (Gen 2) born in Colony Gamma to Parents #10 and #33.

 BIRTH

Tick 7230 (Year 20, Day 30)

Child #66 (Gen 2) born in Colony Alpha to Parents #16 and #24.

 BIRTH

Tick 7340 (Year 20, Day 140)

Child #67 (Gen 2) born in Colony Alpha to Parents #0 and #28.

 BIRTH

Tick 7348 (Year 20, Day 148)

Child #68 (Gen 2) born in Colony Beta to Parents #5 and #29.

 BIRTH

Tick 7348 (Year 20, Day 148)

Child #69 (Gen 2) born in Colony Gamma to Parents #9 and #26.

 BIRTH

Tick 7441 (Year 20, Day 241)

Child #70 (Gen 2) born in Colony Gamma to Parents #8 and #30.

 BIRTH

Tick 7593 (Year 21, Day 33)

Child #71 (Gen 2) born in Colony Alpha to Parents #16 and #24.

 BIRTH

Tick 7643 (Year 21, Day 83)

Child #72 (Gen 2) born in Colony Beta to Parents #4 and #25.

 DEATH

Tick 7674 (Year 21, Day 114)

Agent #43 died at coordinate (135, 515). Primary: Starvation. Secondary: Dehydration.

 DEATH

Tick 7695 (Year 21, Day 135)

Agent #14 died at coordinate (570, 315). Primary: Exposure. Secondary: Injury.

 BIRTH

Tick 7710 (Year 21, Day 150)

Child #73 (Gen 2) born in Colony Beta to Parents #5 and #29.

 BIRTH

Tick 7721 (Year 21, Day 161)

Child #74 (Gen 2) born in Colony Gamma to Parents #9 and #21.

 DEATH

Tick 7754 (Year 21, Day 194)

Agent #12 died at coordinate (552, 304). Primary: Exposure. Secondary: Injury.

 BIRTH

Tick 7764 (Year 21, Day 204)

Child #75 (Gen 2) born in Colony Alpha to Parents #0 and #28.

 BIRTH

Tick 7800 (Year 21, Day 240)

Child #76 (Gen 2) born in Colony Gamma to Parents #10 and #26.

 BIRTH

Tick 7811 (Year 21, Day 251)

Child #77 (Gen 2) born in Colony Gamma to Parents #8 and #30.

 MILESTONE

Tick 7811 (Year 21, Day 251)

New population peak reached: 60 alive at tick 7811!

 BIRTH

Tick 7961 (Year 22, Day 41)

Child #78 (Gen 2) born in Colony Alpha to Parents #16 and #24.

 BIRTH

Tick 8004 (Year 22, Day 84)

Child #79 (Gen 2) born in Colony Beta to Parents #4 and #25.

 DEATH

Tick 8029 (Year 22, Day 109)

Agent #8 died at coordinate (847, 540). Primary: Old Age. Secondary: Injury.

 BIRTH

Tick 8071 (Year 22, Day 151)

Child #80 (Gen 2) born in Colony Beta to Parents #5 and #29.

 BIRTH

Tick 8167 (Year 22, Day 247)

Child #81 (Gen 2) born in Colony Gamma to Parents #20 and #26.

 BIRTH

Tick 8171 (Year 22, Day 251)

Child #82 (Gen 2) born in Colony Alpha to Parents #1 and #28.

 BIRTH

Tick 8179 (Year 22, Day 259)

Child #83 (Gen 2) born in Colony Gamma to Parents #10 and #30.

 DEATH

Tick 8355 (Year 23, Day 75)

Agent #13 died at coordinate (552, 304). Primary: Exposure. Secondary: Injury.

 EXTINCTION

Tick 8400 (Year 23, Day 120)

Colony Delta has gone extinct (all members have died).

 EXTINCTION

Tick 8400 (Year 23, Day 120)

Colony Delta went extinct after surviving for 8400 ticks. No members remain.

 DEATH

Tick 8731 (Year 24, Day 91)

Agent #10 died at coordinate (847, 540). Primary: Old Age. Secondary: Dehydration.

 BIRTH

Tick 8748 (Year 24, Day 108)

Child #84 (Gen 2) born in Colony Alpha to Parents #0 and #28.

 DEATH

Tick 8787 (Year 24, Day 147)

Agent #47 died at coordinate (614, 573). Primary: Injury. Secondary: Dehydration.

 BIRTH

Tick 8850 (Year 24, Day 210)

Child #85 (Gen 2) born in Colony Gamma to Parents #9 and #31.

 BIRTH

Tick 9100 (Year 25, Day 100)

Child #86 (Gen 2) born in Colony Beta to Parents #4 and #37.

 BIRTH

Tick 9113 (Year 25, Day 113)

Child #87 (Gen 2) born in Colony Alpha to Parents #0 and #28.

 DEATH

Tick 9145 (Year 25, Day 145)

Agent #40 died at coordinate (588, 547). Primary: Dehydration. Secondary: Exposure.

 BIRTH

Tick 9215 (Year 25, Day 215)

Child #88 (Gen 2) born in Colony Beta to Parents #29 and #35.

 BIRTH

Tick 9230 (Year 25, Day 230)

Child #89 (Gen 2) born in Colony Gamma to Parents #26 and #31.

 BIRTH

Tick 9296 (Year 25, Day 296)

Child #90 (Gen 2) born in Colony Gamma to Parents #20 and #33.

 BIRTH

Tick 9382 (Year 26, Day 22)

Child #91 (Gen 2) born in Colony Alpha to Parents #16 and #44.

 DEATH

Tick 9411 (Year 26, Day 51)

Agent #81 died at coordinate (833, 534). Primary: Dehydration. Secondary: None.

 DEATH

Tick 9457 (Year 26, Day 97)

Agent #33 died at coordinate (851, 550). Primary: Dehydration. Secondary: Exposure.

 DEATH

Tick 9543 (Year 26, Day 183)

Agent #72 died at coordinate (615, 591). Primary: Dehydration. Secondary: Injury.

 BIRTH

Tick 9574 (Year 26, Day 214)

Child #92 (Gen 2) born in Colony Alpha to Parents #0 and #28.

 BIRTH

Tick 9577 (Year 26, Day 217)

Child #93 (Gen 2) born in Colony Beta to Parents #4 and #29.

 BIRTH

Tick 9609 (Year 26, Day 249)

Child #94 (Gen 2) born in Colony Gamma to Parents #26 and #30.

 BIRTH

Tick 9612 (Year 26, Day 252)

Child #95 (Gen 2) born in Colony Beta to Parents #35 and #42.

 MILESTONE

Tick 9612 (Year 26, Day 252)

New population peak reached: 70 alive at tick 9612!

 BIRTH

Tick 9745 (Year 27, Day 25)

Child #96 (Gen 2) born in Colony Alpha to Parents #16 and #44.

 DEATH

Tick 9827 (Year 27, Day 107)

Agent #27 died at coordinate (839, 540). Primary: Starvation. Secondary: None.

 BIRTH

Tick 9997 (Year 27, Day 277)

Child #97 (Gen 2) born in Colony Beta to Parents #4 and #29.

 MILESTONE

Tick 10000 (Year 27, Day 280)

New longest living lineage: Generation 2 agents are alive!

 DEATH

Tick 10119 (Year 28, Day 39)

Agent #59 died at coordinate (596, 555). Primary: Injury. Secondary: Dehydration.

 BIRTH

Tick 10279 (Year 28, Day 199)

Child #98 (Gen 2) born in Colony Beta to Parents #18 and #35.

 BIRTH

Tick 10359 (Year 28, Day 279)

Child #99 (Gen 2) born in Colony Beta to Parents #4 and #29.

 BIRTH

Tick 10363 (Year 28, Day 283)

Child #100 (Gen 2) born in Colony Alpha to Parents #1 and #28.

 DEATH

Tick 10381 (Year 28, Day 301)

Agent #86 died at coordinate (609, 568). Primary: Dehydration. Secondary: None.

 DEATH

Tick 10435 (Year 28, Day 355)

Agent #76 died at coordinate (843, 537). Primary: Starvation. Secondary: None.

 BIRTH

Tick 10650 (Year 29, Day 210)

Child #101 (Gen 2) born in Colony Alpha to Parents #0 and #46.

 BIRTH

Tick 10720 (Year 29, Day 280)

Child #102 (Gen 2) born in Colony Beta to Parents #29 and #35.

 BIRTH

Tick 10721 (Year 29, Day 281)

Child #103 (Gen 2) born in Colony Beta to Parents #4 and #48.

**BIRTH**

Tick 10732 (Year 29, Day 292)

Child #104 (Gen 2) born in Colony Alpha to Parents #1 and #16.

**BIRTH**

Tick 10845 (Year 30, Day 45)

Child #105 (Gen 2) born in Colony Alpha to Parents #28 and #50.

**BIRTH**

Tick 10942 (Year 30, Day 142)

Child #106 (Gen 3) born in Colony Beta to Parents #51 and #52.

**MILESTONE**

Tick 10942 (Year 30, Day 142)

Generation 3 reached! Max generation in history: 3.

**BIRTH**

Tick 11014 (Year 30, Day 214)

Child #107 (Gen 3) born in Colony Gamma to Parents #31 and #53.

**BIRTH**

Tick 11050 (Year 30, Day 250)

Child #108 (Gen 2) born in Colony Alpha to Parents #0 and #24.

**BIRTH**

Tick 11065 (Year 30, Day 265)

Child #109 (Gen 3) born in Colony Alpha to Parents #46 and #54.

 MILESTONE

Tick 11065 (Year 30, Day 265)

Population recovery — bounced back above 80 (80 alive)!

 MILESTONE

Tick 11065 (Year 30, Day 265)

New population peak reached: 80 alive at tick 11065!

 BIRTH

Tick 11082 (Year 30, Day 282)

Child #110 (Gen 2) born in Colony Beta to Parents #4 and #29.

 BIRTH

Tick 11130 (Year 30, Day 330)

Child #111 (Gen 2) born in Colony Alpha to Parents #1 and #16.

 BIRTH

Tick 11205 (Year 31, Day 45)

Child #112 (Gen 3) born in Colony Gamma to Parents #49 and #55.

 BIRTH

Tick 11253 (Year 31, Day 93)

Child #113 (Gen 2) born in Colony Alpha to Parents #28 and #50.

 BIRTH

Tick 11289 (Year 31, Day 129)

Child #114 (Gen 3) born in Colony Beta to Parents #35 and #56.

 BIRTH

Tick 11305 (Year 31, Day 145)

Child #115 (Gen 3) born in Colony Beta to Parents #51 and #57.

 BIRTH

Tick 11370 (Year 31, Day 210)

Child #116 (Gen 2) born in Colony Beta to Parents #48 and #52.

 BIRTH

Tick 11377 (Year 31, Day 217)

Child #117 (Gen 2) born in Colony Gamma to Parents #20 and #31.

 BIRTH

Tick 11429 (Year 31, Day 269)

Child #118 (Gen 2) born in Colony Alpha to Parents #0 and #46.

 DEATH

Tick 11436 (Year 31, Day 276)

Agent #64 died at coordinate (614, 539). Primary: Dehydration. Secondary: None.

 BIRTH

Tick 11444 (Year 31, Day 284)

Child #119 (Gen 2) born in Colony Beta to Parents #4 and #29.

 BIRTH

Tick 11496 (Year 31, Day 336)

Child #120 (Gen 2) born in Colony Alpha to Parents #1 and #24.

 MILESTONE

Tick 11496 (Year 31, Day 336)

New population peak reached: 90 alive at tick 11496!

 BIRTH

Tick 11614 (Year 32, Day 94)

Child #121 (Gen 3) born in Colony Gamma to Parents #9 and #55.

 BIRTH

Tick 11652 (Year 32, Day 132)

Child #122 (Gen 3) born in Colony Beta to Parents #35 and #56.

 BIRTH

Tick 11709 (Year 32, Day 189)

Child #123 (Gen 3) born in Colony Beta to Parents #51 and #57.

 BIRTH

Tick 11731 (Year 32, Day 211)

Child #124 (Gen 2) born in Colony Beta to Parents #48 and #52.

 BIRTH

Tick 11776 (Year 32, Day 256)

Child #125 (Gen 2) born in Colony Alpha to Parents #28 and #50.

 BIRTH

Tick 11805 (Year 32, Day 285)

Child #126 (Gen 2) born in Colony Beta to Parents #4 and #29.

 BIRTH

Tick 11831 (Year 32, Day 311)

Child #127 (Gen 3) born in Colony Alpha to Parents #46 and #54.

 DEATH

Tick 12332 (Year 34, Day 92)

Agent #90 died at coordinate (837, 539). Primary: Starvation. Secondary: Injury.

 BIRTH

Tick 12402 (Year 34, Day 162)

Child #128 (Gen 3) born in Colony Alpha to Parents #66 and #67.

 BIRTH

Tick 12420 (Year 34, Day 180)

Child #129 (Gen 2) born in Colony Beta to Parents #48 and #57.

 BIRTH

Tick 12483 (Year 34, Day 243)

Child #130 (Gen 3) born in Colony Gamma to Parents #53 and #58.

 BIRTH

Tick 12489 (Year 34, Day 249)

Child #131 (Gen 3) born in Colony Beta to Parents #45 and #51.

 MILESTONE

Tick 12489 (Year 34, Day 249)

Population reached 100! (100 alive)

 MILESTONE

Tick 12489 (Year 34, Day 249)

New population peak reached: 100 alive at tick 12489!

 BIRTH

Tick 12521 (Year 34, Day 281)

Child #132 (Gen 3) born in Colony Gamma to Parents #31 and #49.

 BIRTH

Tick 12680 (Year 35, Day 80)

Child #133 (Gen 2) born in Colony Beta to Parents #4 and #42.

 DEATH

Tick 12706 (Year 35, Day 106)

Agent #52 died at coordinate (589, 548). Primary: Dehydration. Secondary: None.

 BIRTH

Tick 12790 (Year 35, Day 190)

Child #134 (Gen 3) born in Colony Alpha to Parents #66 and #67.

 BIRTH

Tick 12816 (Year 35, Day 216)

Child #135 (Gen 3) born in Colony Alpha to Parents #16 and #75.

 BIRTH

Tick 12822 (Year 35, Day 222)

Child #136 (Gen 2) born in Colony Beta to Parents #29 and #57.

 BIRTH

Tick 12828 (Year 35, Day 228)

Child #137 (Gen 3) born in Colony Beta to Parents #48 and #56.

 BIRTH

Tick 12856 (Year 35, Day 256)

Child #138 (Gen 3) born in Colony Gamma to Parents #21 and #77.

 BIRTH

Tick 12885 (Year 35, Day 285)

Child #139 (Gen 3) born in Colony Beta to Parents #51 and #68.

 DEATH

Tick 12987 (Year 36, Day 27)

Agent #63 died at coordinate (595, 554). Primary: Dehydration. Secondary: None.

 BIRTH

Tick 13052 (Year 36, Day 92)

Child #140 (Gen 3) born in Colony Alpha to Parents #50 and #71.

 BIRTH

Tick 13077 (Year 36, Day 117)

Child #141 (Gen 2) born in Colony Beta to Parents #4 and #37.

 BIRTH

Tick 13096 (Year 36, Day 136)

Child #142 (Gen 3) born in Colony Gamma to Parents #9 and #55.

 BIRTH

Tick 13150 (Year 36, Day 190)

Child #143 (Gen 2) born in Colony Alpha to Parents #41 and #44.

 MILESTONE

Tick 13150 (Year 36, Day 190)

New population peak reached: 110 alive at tick 13150!

 BIRTH

Tick 13155 (Year 36, Day 195)

Child #144 (Gen 3) born in Colony Alpha to Parents #66 and #67.

 BIRTH

Tick 13183 (Year 36, Day 223)

Child #145 (Gen 3) born in Colony Beta to Parents #29 and #80.

 BIRTH

Tick 13199 (Year 36, Day 239)

Child #146 (Gen 3) born in Colony Beta to Parents #35 and #56.

 BIRTH

Tick 13200 (Year 36, Day 240)

Child #147 (Gen 3) born in Colony Alpha to Parents #16 and #75.

 BIRTH

Tick 13219 (Year 36, Day 259)

Child #148 (Gen 2) born in Colony Beta to Parents #48 and #57.

 BIRTH

Tick 13246 (Year 36, Day 286)

Child #149 (Gen 3) born in Colony Beta to Parents #45 and #51.

 BIRTH

Tick 13324 (Year 37, Day 4)

Child #150 (Gen 3) born in Colony Beta to Parents #42 and #68.

 DEATH

Tick 13350 (Year 37, Day 30)

Agent #133 died at coordinate (603, 563). Primary: Dehydration. Secondary: None.

 BIRTH

Tick 13387 (Year 37, Day 67)

Child #151 (Gen 3) born in Colony Gamma to Parents #49 and #62.

 BIRTH

Tick 13589 (Year 37, Day 269)

Child #152 (Gen 2) born in Colony Beta to Parents #4 and #29.

 BIRTH

Tick 13607 (Year 37, Day 287)

Child #153 (Gen 3) born in Colony Beta to Parents #48 and #51.

 DEATH

Tick 13740 (Year 38, Day 60)

Agent #82 died at coordinate (147, 502). Primary: Injury. Secondary: None.

**BIRTH**

Tick 13774 (Year 38, Day 94)

Child #154 (Gen 3) born in Colony Gamma to Parents #9 and #83.

**BIRTH**

Tick 13781 (Year 38, Day 101)

Child #155 (Gen 3) born in Colony Alpha to Parents #28 and #75.

**MILESTONE**

Tick 13781 (Year 38, Day 101)

New population peak reached: 120 alive at tick 13781!

**BIRTH**

Tick 13865 (Year 38, Day 185)

Child #156 (Gen 3) born in Colony Gamma to Parents #20 and #77.

**BIRTH**

Tick 13896 (Year 38, Day 216)

Child #157 (Gen 2) born in Colony Beta to Parents #42 and #57.

**BIRTH**

Tick 13957 (Year 38, Day 277)

Child #158 (Gen 3) born in Colony Beta to Parents #29 and #79.

**BIRTH**

Tick 13962 (Year 38, Day 282)

Child #159 (Gen 2) born in Colony Beta to Parents #35 and #45.

**BIRTH**

Tick 14100 (Year 39, Day 60)

Child #160 (Gen 3) born in Colony Beta to Parents #56 and #80.

**BIRTH**

Tick 14108 (Year 39, Day 68)

Child #161 (Gen 3) born in Colony Alpha to Parents #24 and #61.

**BIRTH**

Tick 14130 (Year 39, Day 90)

Child #162 (Gen 3) born in Colony Beta to Parents #51 and #68.

**BIRTH**

Tick 14148 (Year 39, Day 108)

Child #163 (Gen 3) born in Colony Alpha to Parents #28 and #75.

**BIRTH**

Tick 14166 (Year 39, Day 126)

Child #164 (Gen 3) born in Colony Gamma to Parents #21 and #58.

**BIRTH**

Tick 14179 (Year 39, Day 139)

Child #165 (Gen 3) born in Colony Gamma to Parents #53 and #55.

**MILESTONE**

Tick 14179 (Year 39, Day 139)

New population peak reached: 130 alive at tick 14179!

**BIRTH**

Tick 14257 (Year 39, Day 217)

Child #166 (Gen 2) born in Colony Beta to Parents #48 and #57.

**BIRTH**

Tick 14260 (Year 39, Day 220)

Child #167 (Gen 2) born in Colony Beta to Parents #18 and #42.

**BIRTH**

Tick 14273 (Year 39, Day 233)

Child #168 (Gen 3) born in Colony Gamma to Parents #74 and #89.

**BIRTH**

Tick 14307 (Year 39, Day 267)

Child #169 (Gen 3) born in Colony Alpha to Parents #67 and #78.

**BIRTH**

Tick 14325 (Year 39, Day 285)

Child #170 (Gen 3) born in Colony Beta to Parents #35 and #88.

**BIRTH**

Tick 14461 (Year 40, Day 61)

Child #171 (Gen 3) born in Colony Beta to Parents #56 and #80.

**BIRTH**

Tick 14472 (Year 40, Day 72)

Child #172 (Gen 3) born in Colony Alpha to Parents #24 and #61.

**BIRTH**

Tick 14491 (Year 40, Day 91)

Child #173 (Gen 3) born in Colony Beta to Parents #51 and #68.

**BIRTH**

Tick 14516 (Year 40, Day 116)

Child #174 (Gen 3) born in Colony Alpha to Parents #28 and #75.

**BIRTH**

Tick 14625 (Year 40, Day 225)

Child #175 (Gen 2) born in Colony Beta to Parents #42 and #48.

**MILESTONE**

Tick 14625 (Year 40, Day 225)

New population peak reached: 140 alive at tick 14625!

**BIRTH**

Tick 14659 (Year 40, Day 259)

Child #176 (Gen 3) born in Colony Gamma to Parents #55 and #77.

**BIRTH**

Tick 14672 (Year 40, Day 272)

Child #177 (Gen 3) born in Colony Alpha to Parents #78 and #91.

**BIRTH**

Tick 14711 (Year 40, Day 311)

Child #178 (Gen 3) born in Colony Beta to Parents #35 and #88.

 BIRTH

Tick 14792 (Year 41, Day 32)

Child #179 (Gen 3) born in Colony Alpha to Parents #41 and #67.

 BIRTH

Tick 14822 (Year 41, Day 62)

Child #180 (Gen 3) born in Colony Beta to Parents #56 and #80.

 BIRTH

Tick 14838 (Year 41, Day 78)

Child #181 (Gen 3) born in Colony Alpha to Parents #84 and #96.

 BIRTH

Tick 14838 (Year 41, Day 78)

Child #182 (Gen 3) born in Colony Beta to Parents #93 and #95.

 BIRTH

Tick 14852 (Year 41, Day 92)

Child #183 (Gen 3) born in Colony Beta to Parents #45 and #51.

 BIRTH

Tick 14889 (Year 41, Day 129)

Child #184 (Gen 2) born in Colony Alpha to Parents #24 and #28.

 DEATH

Tick 14946 (Year 41, Day 186)

Agent #9 died at coordinate (846, 540). Primary: Old Age. Secondary: Dehydration.

**BIRTH**

Tick 14970 (Year 41, Day 210)

Child #185 (Gen 3) born in Colony Beta to Parents #57 and #68.

**BIRTH**

Tick 15014 (Year 41, Day 254)

Child #186 (Gen 2) born in Colony Beta to Parents #37 and #42.

**MILESTONE**

Tick 15014 (Year 41, Day 254)

New population peak reached: 150 alive at tick 15014!

**BIRTH**

Tick 15030 (Year 41, Day 270)

Child #187 (Gen 3) born in Colony Gamma to Parents #49 and #55.

**BIRTH**

Tick 15036 (Year 41, Day 276)

Child #188 (Gen 3) born in Colony Alpha to Parents #75 and #78.

**BIRTH**

Tick 15037 (Year 41, Day 277)

Child #189 (Gen 3) born in Colony Beta to Parents #48 and #97.

**BIRTH**

Tick 15113 (Year 41, Day 353)

Child #190 (Gen 3) born in Colony Alpha to Parents #50 and #91.

 DEATH

Tick 15134 (Year 42, Day 14)

Agent #129 died at coordinate (583, 543). Primary: Starvation. Secondary: Dehydration.

 BIRTH

Tick 15183 (Year 42, Day 63)

Child #191 (Gen 3) born in Colony Beta to Parents #35 and #56.

 BIRTH

Tick 15199 (Year 42, Day 79)

Child #192 (Gen 3) born in Colony Beta to Parents #80 and #93.

 BIRTH

Tick 15204 (Year 42, Day 84)

Child #193 (Gen 3) born in Colony Alpha to Parents #41 and #96.

 BIRTH

Tick 15213 (Year 42, Day 93)

Child #194 (Gen 3) born in Colony Beta to Parents #45 and #51.

 BIRTH

Tick 15247 (Year 42, Day 127)

Child #195 (Gen 3) born in Colony Beta to Parents #79 and #95.

 BIRTH

Tick 15253 (Year 42, Day 133)

Child #196 (Gen 2) born in Colony Alpha to Parents #24 and #28.

 DEATH

Tick 15269 (Year 42, Day 149)

Agent #103 died at coordinate (640, 567). Primary: Dehydration. Secondary: Exposure.

 BIRTH

Tick 15272 (Year 42, Day 152)

Child #197 (Gen 3) born in Colony Gamma to Parents #21 and #89.

 BIRTH

Tick 15328 (Year 42, Day 208)

Child #198 (Gen 3) born in Colony Beta to Parents #73 and #98.

 MILESTONE

Tick 15328 (Year 42, Day 208)

New population peak reached: 160 alive at tick 15328!

 BIRTH

Tick 15331 (Year 42, Day 211)

Child #199 (Gen 3) born in Colony Beta to Parents #57 and #68.

 BIRTH

Tick 15367 (Year 42, Day 247)

Child #200 (Gen 3) born in Colony Alpha to Parents #61 and #67.

 EXTINCTION

Tick 36100 (Year 100, Day 100)

Colony Delta went extinct after surviving for 36100 ticks. No members remain.



Timeline Explorer & History Scrubber

Civilization Time Machine

Analyze demographic shifts, emotional climates, and selective pressures. Use the slider playhead or click any milestone anchor to jump the visualizer to that tick.

TIMELINE PLAYHEAD: Tick 0

MILESTONE ANCHORS & DISASTER JUMPS

Layer 1 — Facts & Observations

POPULATION
0

CLIMATE EPOCH
Temperate

STORED RESOURCES
F:0 | W:0

CUMULATIVE BIRTHS
0

CUMULATIVE DEATHS
0

DOMINANT EMOTION
Neutral

AVG NOVELTY
0.00

AVG TRUST
0.00

MAJOR HISTORICAL EVENT (TICK WINDOW)
None

Dual Point Compare (Overlay Mode)

Select a historical compare target point to calculate relative delta changes.

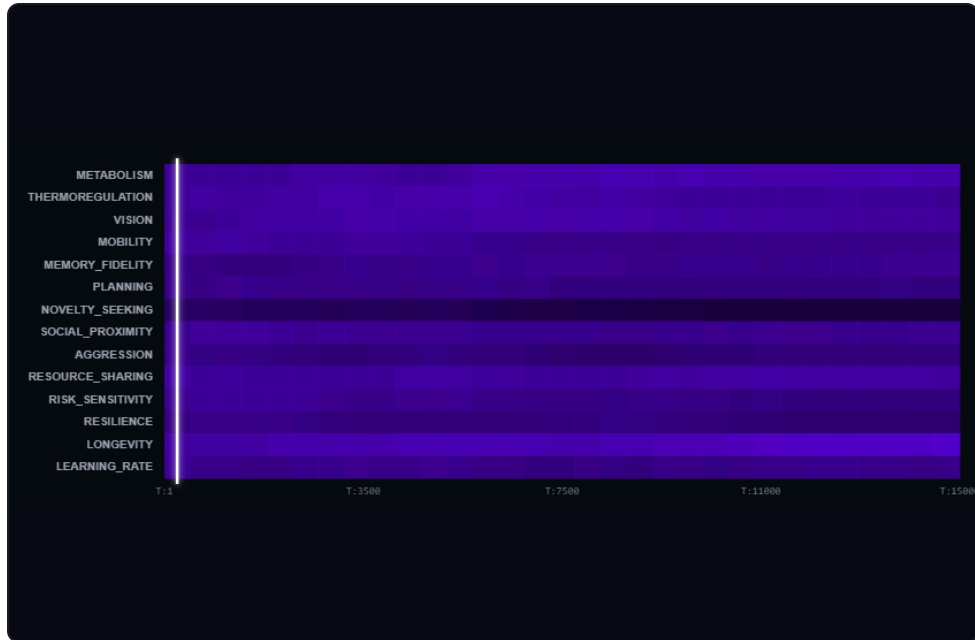
METRIC	CURRENT (BASE)	TARGET (COMPARE)	DELTA CHANGE
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Layer 3 — Scientific Hypotheses (Auto-Generated)

Statistical causal hypotheses generated by comparing consecutive climate transitions. Click a card to expand and highlight affected metrics on charts.

Genotype Evolution Heatmap

Heatmap represents average gene pool parameters of living agents over time. Ticks on X-axis, Genes on Y-axis. Vertical white line shows the current scrubber playhead position.





Experiment Comparison Workbench

Load Secondary Experiment

Compare the current simulation run side-by-side with a secondary experiment run. Select a run from the history catalog dropdown or load a local file.

or select from history: